Empório da Música — AI Customer Support Agent

Technical Challenge — AI Engineer (Artefact) | Project README & Architecture Documentation

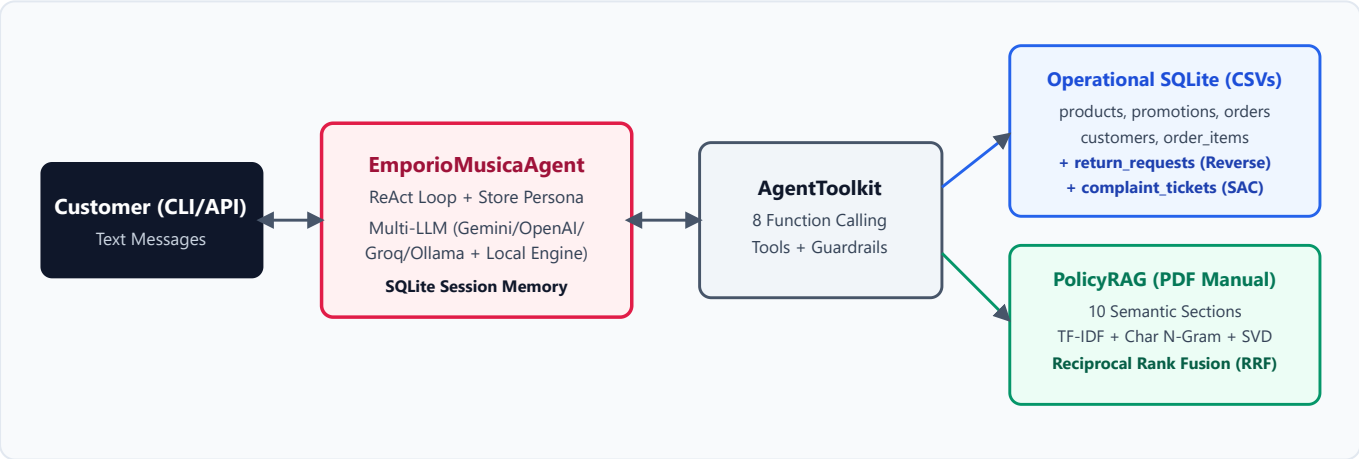
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Stack: Python 3.10+ | SQLite | Hybrid RAG | ReAct

🔗 **Public GitHub Repository:** https://github.com/GaabrielCH/desafio-ai-engineer-artefact-emporio-musica
Includes modular source code (`src/`), automated test suite (`tests/`), 5 full multi-turn conversation transcripts (`examples/`), and a progressive Git commit history.

1. Overview & System Architecture

Empório da Música (a musical instrument store located in Campo Grande/MS, Brazil) requires an AI assistant capable of handling two fundamentally different information retrieval paradigms: **(1) Structured Transactional Data (`data/*.csv`)** — product catalog, live stock levels, pricing, active vs. expired promotions, and customer orders — which require zero numerical hallucination; and **(2) Unstructured Normative Policies (`data/políticas_da_loja.pdf`)** — return/exchange rules, consumer warranty (CDC), shipping modalities, business hours, and store scope guardrails.

To solve both requirements with production-grade reliability, we implemented a **Hybrid Agent Architecture combining Function Calling (ReAct) + Sanitized Relational SQLite Store + Tri-Modal Policy RAG with Reciprocal Rank Fusion (RRF)**:



2. Technical Decisions & Justification

Decision Area	Chosen Approach	Technical Justification
Agent Architecture	**Hybrid: Function Calling (ReAct) + Relational SQL + Policy RAG**	Tabular data (`products.csv` , `orders.csv` , `promotions.csv`) requires exact numerical filtering (`price ≤ 1000` , `stock > 0` , `is_active == 1`) and relational joins, where pure vector RAG would hallucinate figures. Conversely, the PDF policy manual is best served by structural section-based RAG.
Policy Retrieval (PDF RAG)	**Section Chunking + Tri-Modal (Word TF-IDF + Char N-Grams + LSA/SVD Dense) + RRF**	The 8-page PDF is segmented into its 10 numbered semantic sections (preserving payment/shipping tables intact). Retrieval fuses lexical word n-grams, character n-grams (typo-resilient), and L2-normalized dense semantic vectors (`TruncatedSVD`) via *Reciprocal Rank Fusion (RRF)*.
Deterministic Business Logic	**8 Specialized Python Tools with Built-in Calculators**	Critical financial and legal rules — 5% PIX discount non-cumulativity with active promos (Sec. 6.2), tiered interest-free installments (min. R$ 50 for 1–3x; R$ 80 for 4–6x; R$ 100 for 7–12x — Sec. 3.1), and calendar-day windows for Buyer's Remorse (7 days) and Warranty (30/90 days) — are computed deterministically in Python.
LLM Provider & Model	**Multi-Provider (Gemini, OpenAI, Groq, Ollama) + Local ReAct Fallback**	Supports native API Function Calling (`gemini-2.5-flash` , `gpt-4o-mini` , `llama-3.3-70b`) and includes a zero-config deterministic local ReAct engine (`LLM_PROVIDER=auto`) so evaluators can run and test the entire system out-of-the-box after `git clone` even without external API keys.
Conversation Persistence	**SQLite (`data/chat_history.db`)**	Persists sessions, user/assistant turns, and full JSON tool call traces, enabling multi-turn coreference resolution (e.g., "*What about its installments?*") and auditability.

3. Data Sanitization, Detected Anomalies & Documented Assumptions

During exploratory data analysis of `data/*.csv` and `data/políticas_da_loja.pdf` , we identified and programmatically handled (`src/database.py` , `src/policy_rag.py`) the following edge cases:

- **Active Product with Zero Stock (`out_of_stock`):** Acoustic guitar *Giannini GF-3D Dreadnought Sunburst* (`product_id=96`) has `status='active'` but `stock_quantity=0` . Per Section 7.3, it is excluded from available searches and, when queried directly, the tool flags unavailability and automatically recommends 3 in-stock alternatives.
- **Discontinued Products with Residual Stock (`discontinued`):** Bass *Ibanez Bass 2X* (`product_id=136`) has `stock_quantity=13` but `status='discontinued'` (along with 7-string guitar *Shelby SN-7C*, ID 113). The `discontinued` status takes precedence and the agent suggests active equivalents (Section 7.3).
- **Active vs. Expired Promotions & PIX Non-Cumulativity:** Out of 25 rows in `promotions.csv` , only **4 are active (`is_active=1`)**: *Kalani KAL-700T* (-10%), *Ohana CK-20* (-20%), *Crafter HT-100* (-18%), and *Taylor 110e* (-8%). The remaining 21 (including *Takamine GD20* at 12%) are expired (`is_active=0`). The agent transparently informs when a past promotion has ended (Section 7.3) and prevents stacking the 5% PIX discount on already discounted items (Section 6.2).
- **Title vs. Description Mismatches (IDs 135–144) & Empty Categories (IDs 6, 7, 8):** Synthetic items where the description mentions a different brand than the title are flagged with a `data_anomaly` prioritizing the canonical `name` . Categories *Wind Instruments (Woodwinds/Brass)* and *Orchestral Strings* (0 rows in `products.csv`) return a transparent stock notice.

- **PDF Internal Ambiguities Reconciled:** (a) Section 1.2 lists phone/WhatsApp `(67) 3341-4444` while Section 7 lists `(67) 3321-4500`; the agent provides both (physical store vs. remote WhatsApp central). (b) Table 3 summarizes a R$ 100 minimum installment, while Section 3.1 details the exact tiers: R$ 50 (1–3x), R$ 80 (4–6x), and R$ 100 (7–12x), which we enforced in the payment calculator.

4. Production Evolutions Implemented Beyond Base Scope

🔒 1. LGPD 2-Factor Verification & PII Masking

Compliant with Section 9 (LGPD): querying an order by ID alone (e.g., `#7`) masks customer PII (`le**a@jmail.com`, `(67) *****-0123`). Providing a matching 2nd factor (name, email, or phone) unlocks `verificado_2_fatores`.

🚚 2. Weight/Cubage & Regional Tariff Shipping Engine

Estimates package weight/dimensions by category (Ukulele 1.2 kg; Acoustic Guitar 3.2 kg; Electric Guitar 4.8 kg; Drums 28 kg) + regional tariff calculation (MS Interior vs. Other States) + 1.5% *ad valorem* insurance for PAC, Jadlog, and SEDEX.

🎫 3. Reverse Logistics & SAC Escalation Write Tools

Transactional tools `registrar_solicitacao_troca_devolucao` (issues protocol `TROCA-2026-XXXX` and prepaid Correios reverse code `LOG-REV-BR ...`) and `abrir_chamado_reclamacao` (protocol `SAC-2026-XXXX` with 24-business-hour SLA — Sec. 7.3).

🧠 4. Dense Semantic RAG (LSA/SVD) + RRF

Upgraded `PolicyRAG` combining 3 retrieval signals (sublinear word TF-IDF, char_wb 3..5 n-grams, and L2-normalized `TruncatedSVD` dense semantic embeddings) fused via *Reciprocal Rank Fusion (RRF)*.

5. Setup & Execution Instructions

```
# 1. Clone the repository and install dependencies
git clone https://github.com/GaabrielCH/desafio-ai-engineer-artefact-emporio-musica.git
cd desafio-ai-engineer-artefact-emporio-musica
pip install -r requirements.txt

# 2. Configure environment variables (.env is optional: works out-of-the-box with the local ReAct
#    engine, or connects natively to GEMINI_API_KEY, OPENAI_API_KEY, GROQ_API_KEY,
OPENAI_BASE_URL)
cp .env.example .env

# 3. Start the interactive CLI chat or run the automated 5-scenario terminal demonstration
python main.py
python main.py --demo

# 4. Run the automated test suite (8 unit & integration tests via pytest)
python -m pytest -v
```

6. Conversation Examples Included in the Repository (`examples/`)

📄 01_catalogo_violoes_ate_1000.md — Catalog Search & Out-of-Stock / Discontinued Handling

Lists the 12 available acoustic guitars up to R$ 1,000.00 (grouped into nylon and steel strings with PIX and installment pricing). In Turn 2, handles queries for *Giannini GF-3D* (0 stock) and *Shelby SN-7C* (discontinued), applying Section 7.3 and recommending in-stock alternatives.

📄 02_preco_promocoes_e_pagamento_takamine.md — Pricing, Expired vs. Active Promotions & PIX Rules

Details the *Takamine GD20* (R$ 2,199.00 | PIX R$ 2,089.05 | 12x of R$ 183.25 | combined payment > R$ 2,000), transparently informs that its past 12% promo expired (`is_active=0`), and explains PIX non-cumulativity on the actively discounted *Taylor 110e* (-8%).

📄 **03_devolucao_arrependimento_e_rastreio_pedidos.md — Non-Trivial Scenario: CDC Policies + Live Order + Reverse Logistics**

Explains Section 4 return rules, verifies **Order #7** for customer *Letícia Gonçalves Rocha* (2FA LGPD verification), confirms eligibility within the 7-calendar-day delivery window, automatically issues protocol `TROCA-2026-0007` + prepaid reverse code `LOG-REV-BR0007EMP26BR`, and tracks **Order #8** (`BRJL5544332BR`).

📄 **04_informacoes_gerais_frete_e_horario.md — Store Info, Opening Hours & Regional Shipping Quotes**

Provides store address, hours (Mon–Fri 09:00–18:00, Sat 09:00–13:00), and compares shipping for a R$ 600 guitar to **Campo Grande/MS** (Free > R$ 500 via own courier in 1–3 business days) vs. **Dourados/MS** (PAC R$ 55.10, Jadlog R$ 67.34, SEDEX R$ 87.46 with insurance included).

📄 **05_fora_de_escopo_acessorios_e_off_topic.md — Scope Guardrails (Accessories & Off-Topic Requests)**

Politely declines accessory requests (Elixir strings, picks, and guitar amplifier — Sections 1 & 7.1, as the store sells exclusively musical instruments) and redirects off-topic questions (carrot cake recipe) while maintaining the store persona.

7. Known Limitations, Next Steps & AI Assistant Workflow

- **Known Limitations & Production Next Steps:** (1) Bind the 2FA LGPD verification automatically to the authenticated WhatsApp Business Cloud API phone token or SMS/Email OTP; (2) Connect shipping and reverse-logistics tools to live Correios/Melhor Envio REST APIs and ERP inventory webhooks; (3) Expose a web dashboard for human agents to manage the 24-hour SLA queue in `complaint_tickets`.
- **AI Coding Assistant Workflow:** Developed using **Antigravity (Google DeepMind)** in an agentic pair-programming workflow: (1) automated exploratory audit of all CSVs and OCR of the PDF manual to map data anomalies prior to coding; (2) architectural separation between relational SQLite tools and section-based Policy RAG; (3) modular implementation validated by 8 automated `pytest` tests; and (4) reproducible generation of the 5 conversation transcripts in `examples/`.